



HUMBOLDT-UNIVERSITÄT ZU BERLIN
BERNSTEIN CENTER FOR
COMPUTATIONAL NEUROSCIENCE



Models of Neural Systems, WS 2024/25
Project 1: Phase-Locking of Local Oscillations

Supervised by: Stefano Masserini, R. Kempter's lab

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Introduction Brain oscillations are typically categorized based on the frequency band to which they belong. Gamma oscillations (30–100 Hz) can emerge by a few different mechanisms, the most prominent being the pyramidal–interneuronal network gamma (PING) oscillation, in which pyramidal cells (excitatory, E) and interneurons (inhibitory, I) interact with each other in a periodic loop of excitation and inhibition. Although this oscillation is usually generated locally (just a few neurons of the two kinds are sufficient to generate it), oscillations at a similar frequency tend to entrain and to oscillate with the same frequency and a fixed lag. This phenomenon, called phase-locking, can occur with a wide-variety of lags (Lowet et al., 2015). Depending on the delay that spikes need to travel between the two local circuits, there is an optimal lag (which is not always the zero-lag, or in-phase synchrony) which makes one of the circuits most responsive to the other one. This is because local oscillatory circuits are more responsive to inputs which arrive in their more excitable phase, according to the communication through coherence hypothesis (Fries, 2005).

Dumont and Gutkin (2019) have researched the question of how the lag of the phase-locking depends on crucial parameters such as the transmission delay and the strength of different cross-connections between the two local networks. In their study, they combine simulation of spiking networks with a refined analytical approach based on the phase-response curve of an equivalent mean-field model. In this project, the aim is to replicate the spiking network part.

Steps

1. Read Keeley et al. (2017) and familiarize yourself with the context, methods, and results of this paper. Focus primarily on the spiking model and on the PING oscillation, since the rate model and the ING oscillation are not included in this project.
2. Using Brian (www.briansimulator.org), implement the network of quadratic integrate-and-fire neurons described in the beginning of the Results (Equations 1 and following, until Equation 2 excluded). Note that the values of the parameters are rather inconveniently written in the caption of each figure.
3. Simulate one E–I network with the parameters in Figure 2 and see whether you also observe a PING oscillation like in panel D. Explore the effects of varying relevant parameters, like the external input to population E (but potentially also parameters like recurrent excitation and recurrent inhibition, which in the paper are left to 0).

4. Couple two E–I networks and think of a good way of estimating their lag. Then explore how the lag varies as a function of delay, cross–E→E and cross–E→I connection, and compare to Figures 5 and 6. In addition, you can investigate the effect of the cross–I→E and cross–I→I connections, which were not considered in the paper.
5. **(Extra)** If the simulation time is not too long, you may try to reproduce the phase-response curve in the local network (Figure 2E) and in the coupled one (Figure 5). Note that the latter was done for the mean-field model and that doing it in the spiking model might be quite computationally expensive.

Hints

- Note that the phase-lag plots in Figures 5–6 are estimated analytically for the mean-field model, so they will necessarily look more noisy for the spiking network. However, they should have the same shape.
- For the same reason, replicating those plots for the spiking network might be computationally demanding. In this case, some possibilities are: 1) to scale down the network (e.g. only 1000 or 500 E and I neurons) and 2) to sample just enough parameter values to get an idea of the shape, without a lot of resolution. The same hints apply for the phase-response curves (extra task).

Create labeled figures for all your main findings and summarize your results in a brief report.

References

- G. Dumont and B. Gutkin. Macroscopic phase resetting-curves determine oscillatory coherence and signal transfer in inter-coupled neural circuits. *PLoS computational biology*, 15(5):e1007019, 2019.
- P. Fries. A mechanism for cognitive dynamics: neuronal communication through neuronal coherence. *Trends in cognitive sciences*, 9(10):474–480, 2005.
- S. Keeley, A. A. Fenton, and J. Rinzel. **Modeling fast and slow gamma oscillations with interneurons of different subtype**. *Journal of neurophysiology*, 117(3):950–965, 2017.
- E. Lowet, M. Roberts, A. Hadjipapas, A. Peter, J. van der Eerden, and P. De Weerd. Input-dependent frequency modulation of cortical gamma oscillations shapes spatial synchronization and enables phase coding. *PLoS computational biology*, 11(2):e1004072, 2015.